॥ श्रीगणेशाय नमः ॥

Greedy Algorithms

Fractional knapsack

We've been given a knapsack with a certain capacity along with some items (each with associated profits and weights). We need to choose items to put into the knapsack such that the total profit is maximized.

	object1	object2	object3
profits	25	24	15
weights	18	15	10

Say capacity = 20

A greedy algorithm could be greedy about the profits.

Say we choose obj 1 : space_left = capacity - wt. of obj 1 = 20 - 18 = 2

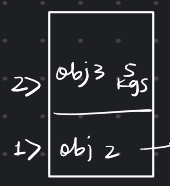
obj 2 can't fit in fully.

A greedy algorithm could be greedy about the weights. It fits in as many items as it can by choosing the lightest objects.



A greedy algorithm could be greedy about the unit cost/ unit profit. It'll maximize the profit/weight ratio.

	object1	object2	object3
profits	25	24	15
weights	18	15	10
P/W	1.4	1.6	1.5



$P = \frac{5}{10} \times 15 = 7.5$

$P = 24$

Total profit = 31.5

The maximum profit/ratio gives the best results.

capacity = 15							
objects	1	2	3	4	5	6	7
profits	10	5	15	7	6	18	3
weights	2	3	5	7	1	4	1

P/W 5 1.66 3 1 6 4.5 3

capacity = 15.

obj 5 : left = 14 profit = 6

obj 1 : left = 12 profit = 16

obj 6 : left = 12-4 = 8 profit = 34

obj. 3 : left = 8-5 = 3 profit = 49

obj. 7 : left = 3-1 = 2 profit = 52

obj. 2 (partial) : left = 2-2 = 0 Profit = $52 + \frac{2}{3} \times 5 = 55.3$

```
GREEDY_KNAPSACK():
    for i=1 to n:
        compute pi/wi
    sort the items in non-increasing order of pi/wi

    for i=1 to n:
        if m > 0 and wi <= m:
            m = m - wi
            profit = profit + pi
            xi = 1
        else:
            break;

        if m > 0:
            profit = profit + (m/wi)*pi
            m = 0
            break;
```

capacity = 15					
objects	1	2	3	4	5
profits	2	28	25	18	9
weights	1	4	5	3	3

P/W 2 7 5 6 3

Capacity = 15

obj. 2 : left = 15-4 = 11 profit = 28

obj. 4 : left = 11-3 = 8 profit = 28+18 = 46

obj. 3 : left = 8-5 = 3 Profit = 46+25 = 71

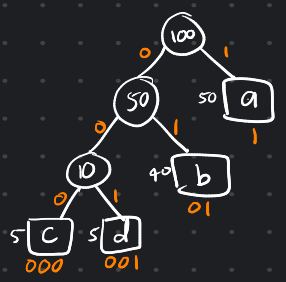
obj. 5 : left = 3-3 = 0 profit = 71+9 = 80

max profit = 80

Huffman coding

Transmitting characters would be costlier than if we use some variable-sized numeric encoding. Huffman coding is used to encode characters to bits.

character	a	b	c	d
frequency	50	40	5	5



a - 1
b - 01
c - 001
d - 000

weighted external path length = $5\times3+5\times3+40\times2+50\times1$

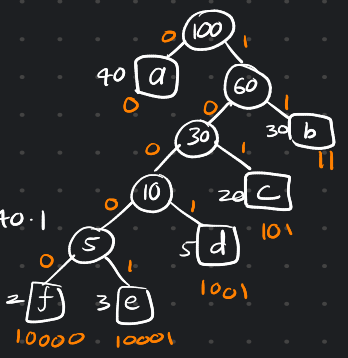
Q. What is the weighted external path length?

character	a	b	c	d	e	f
frequency	40	30	20	5	3	2

→ w.e.p.l =

$2\cdot5+3\cdot5+5\cdot4+20\cdot3+30\cdot2+40\cdot1$

$= 205$



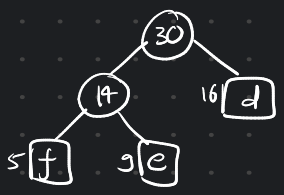
Q. What is the weighted external path length?

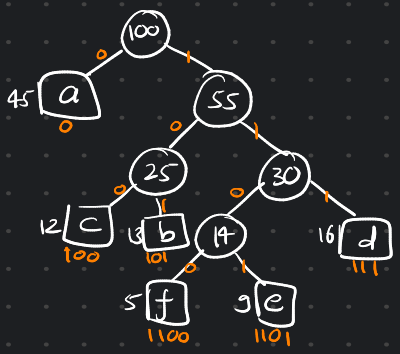
character	a	b	c	d	e	f
frequency	45	13	12	16	9	5



45, 16, 14, 25

45, 30, 25





$45\cdot1+12\cdot3+13\cdot3+16\cdot3+5\cdot4+9\cdot4=224$

```
HUFFMAN(c):
    n = len(c)
    make a min heap Q with c ←O(n)
    for i=1 to n-1:
        allocate a new node z
        z.left = x = extract_min(Q)
        z.right = y = extract_min(Q)
        z.freq = x.freq + y.freq
        insert(Q, z)←O(logn)
    return extract_min(Q)
```

→ O(log n)

← O(n log n)

1.12.2 Huffman Code: GATE CSE 2007 | Question: 76

Suppose the letters a, b, c, d, e, f have probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{32}$, respectively.

Which of the following is the Huffman code for the letter a, b, c, d, e, f?

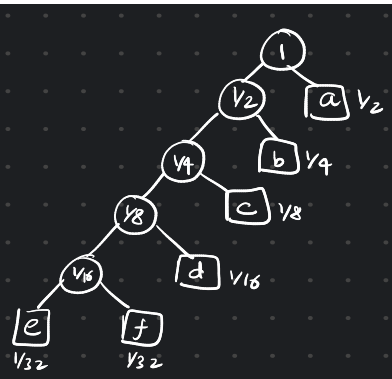
A. $0, 10, 110, 1110, 11110, 11111$

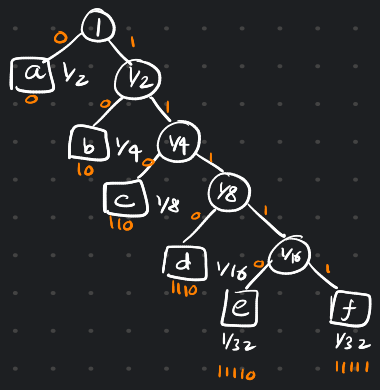
B. $11, 10, 011, 010, 001, 000$

C. $11, 10, 01, 001, 0001, 0000$

D. $110, 100, 010, 000, 001, 111$

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Q. What is the average bit-length per char of the codes for ans in the prev question?

$$\frac{\frac{1}{2}\cdot 1 + \frac{1}{4}\cdot 2 + \frac{1}{8}\cdot 3 + \frac{1}{16}\cdot 4 + \frac{1}{32}\cdot 5 + \frac{1}{32}\cdot 5}{\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{32}}$$

1.12.3 Huffman Code: GATE CSE 2007 | Question: 77

Suppose the letters a, b, c, d, e, f have probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{32}$, respectively.

What is the average length of the Huffman code for the letters a, b, c, d, e, f?

A. 3

B. 2.1875

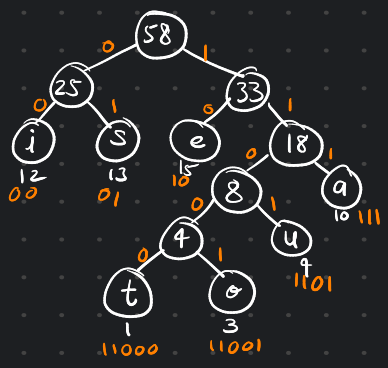
C. 2.25

D. 1.9375

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Q. What is the weighted external path length?

a	e	i	o	u	s	t
10	15	12	3	4	13	1



$$12\cdot 2 + 13\cdot 2 + 15\cdot 2 + 10\cdot 3 + 4\cdot 4 + 3\cdot 5 + 1\cdot 5$$

$$= 146$$

Job sequencing with deadlines

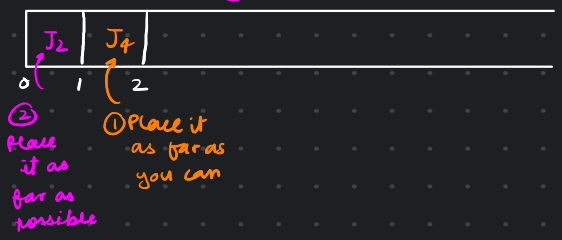
Pick the job with the highest profit and place it as far as possible.

	J1	J2	J3	J4
Deadlines	2	1	1	2
Profits	6	8	5	10

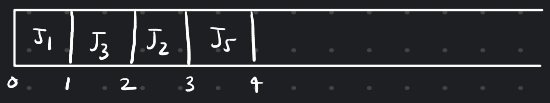
① First take this

② Place it as far as possible

① Place it as far as you can



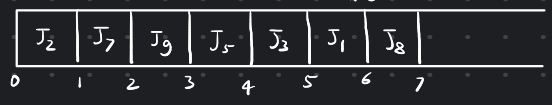
	J1	J2	J3	J4	J5
Profits	2	4	3	1	10
Deadlines	3	3	3	4	4



Profit = 2+3+4+10 = 19

	J1	J2	J3	J4	J5	J6	J7	J8	J9
Profits	15	20	30	18	18	10	23	16	25
Deadlines	7	2	5	3	4	5	2	7	3

CAN'T TAKE



profit = 15+20+30+18+23+16+25
= 40+50+34+23
= 90+57
= 147

ALGORITHM

Sort the jobs accd to profit. ← O(nlogn)

For each job:

Go to deadline and scan backwards for vacant slot.] → $O(n^2)$

Optimal merge patterns

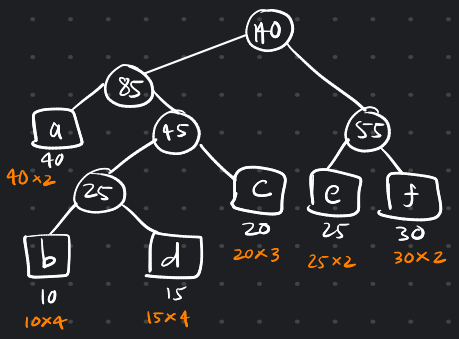
Say we've got separate files with sorted records in them. We want to merge them in a way to minimize the external weighted path length.

a	b	c	d	e	f
40	10	20	15	25	30

We need to find the # of record movements.

Q. Total # of record movements?

→



Total # of movements = $40\times2 + 10\times4 + 15\times4 + 20\times3 + 25\times2 + 30\times2$